Definition: A function is **continuous at** $x = a$ if $\lim_{x \to a} f(x) = f(a)$.

So the definition requires the following three things in order for f to be continuous at $x = a$.

1.) $f(a)$ is defined.

2.) $\lim_{x \to a} f(x)$ exists

3.) $\lim_{x \to a} f(x) = f(a)$

Example: Use the **definition of continuity at a number** to explain why the function is discontinuous at the given number a. Use appropriate limit notation.

$$f(x) = \begin{cases} 7x - 1 & if\ x \leq 3 \\ x^2 + 2 & if\ x > 3 \end{cases} \qquad a = 3$$

f is discontinuous at $a = 3$ because $\lim_{x \to 3} f(x)$ does not exist.

$$\lim_{x \to 3^+} f(x) = \lim_{x \to 3^+} (x^2 + 2) = 3^2 + 2 = 11$$

$$\lim_{x \to 3^-} f(x) = \lim_{x \to 3^-} (7x - 1) = 7(3) - 1 = 20$$

$$\lim_{x \to 3^+} f(x) = 11 \neq 20 = \lim_{x \to 3^-} f(x) \Rightarrow \lim_{x \to 3} f(x) \text{ does not exist}$$